

Problem B. Beyond Journey's End

Input file: stdin
Output file: stdout
Time limit: 1 second
Memory limit: 256MB

Statement

After the Hero's party defeated the Demon King, the continent gradually returned to peace.

The elf mage **Frieren** set out on a journey to collect spells.

Along the way, villages may post two kinds of commissions:

- **Support commissions** (e.g., helping a village, assisting a cleric), which provide **positive reward** (such as receiving spellbooks).
- **Suppression commissions** (e.g., guarding a village, repelling demons), which require **workload** to complete and also provide **reward**.

Each commission event occurs at a village (a node) and lasts over a time interval $[L, R)$.

The timeline always starts at time 0.

Since commission availability changes only at event endpoints, you may divide the timeline into **time segments** induced by all distinct endpoints of the given intervals together with 0.

Let these sorted distinct time points be:

$$0 = t_0 < t_1 < \cdots < t_M.$$

They define M time segments $[t_k, t_{k+1})$. Within a segment, the set of active events is unchanged.

In every time segment, Frieren may either:

1. **Work** at a chosen village, gaining reward according to the active commissions there, or
2. **Rest** at her current village, gaining reward 0 for that segment.

Resting means Frieren **does not move during that segment** and stays at the same village as in the previous segment.

For a time segment k and village v :

- $E_k(v)$ is the total intensity of **support commissions** at village v during segment k .
- $L_k(v)$ is the total workload of **suppression commissions** at village v during segment k .
- $T_k = t_{k+1} - t_k$ is the duration of segment k .

However, casting demon-slaying magic *Zoltraak* consumes mana and resources, so only a fraction of the suppression workload is converted into effective reward. Given an integer percentage C ($0 < C \leq 100$), during segment k at village v , the suppression part contributing to reward is

$$\left\lfloor \frac{C \cdot L_k(v)}{100} \right\rfloor$$

Therefore, if she chooses to work at village v during segment k , her reward is

$$R_k(v) = T_k \cdot \left(E_k(v) + \left\lfloor \frac{C \cdot L_k(v)}{100} \right\rfloor \right).$$

If she rests during segment k , her reward for that segment is 0.

The continent map is an undirected weighted graph with N nodes (villages).

Frieren possesses a powerful fast-travel spell that allows her to move between villages instantly without consuming any time. However, casting this spell consumes mana proportional to the travel distance.

Therefore, although moving between villages does not reduce the duration of any time segment, it reduces her total reward by an amount equal to the **shortest-path distance** traveled.

Let S be the starting village at time 0. If Frieren's village sequence over all segments is

$$(v_0, v_1, \dots, v_{M-1}),$$

then her total movement cost is

$$\text{dist}(S, v_0) + \sum_{k=1}^{M-1} \text{dist}(v_{k-1}, v_k).$$

A transition between two disconnected villages is invalid.

Frieren wants to maximize:

(total reward) – (total movement cost).

Your task is to compute this maximum value.

Input

N E Q S
 u v w (E lines)
 L R pos p (Q lines)
 C

- N : number of villages ($1 \leq N \leq 2000$)
- E : number of edges ($0 \leq E \leq 5000$)
- Each edge is undirected with weight w ($1 \leq w \leq 10^6$)
- Q : number of commission events ($0 \leq Q \leq 2 \times 10^5$)

- S : starting village (1-indexed)
- $0 \leq L < R \leq 10^9$
- $|p| \leq 10^6$
- $0 < C \leq 100$

Let K be the number of distinct time points in the set $\{0\} \cup \{L_i, R_i\}_{i=1}^Q$.

It is guaranteed that $K \leq 2000$.

Each event (L, R, pos, p) describes a commission active at village `pos` during $[L, R)$:

- If $p > 0$: it contributes p to **support intensity**.
- If $p < 0$: it contributes $|p|$ to **suppression workload**.

If $Q = 0$, the timeline contains a single segment $[0, 1)$.

Output

Print a single integer — the maximum value of

(total reward – total movement cost).

Negative values are allowed.

Example 1

stdin	stdout
4 4 4 1 1 2 3 2 3 2 3 4 4 1 4 10 0 2 2 5 1 3 3 -2 2 4 4 4 3 5 1 -1 100	9

Example 2

stdin	stdout
3 2 4 1 1 2 3 2 3 3 0 1 1 -2 1 4 2 5 1 4 2 4 1 4 2 -6 50	34

Explanation

- Distinct time points: $0, 1, 4 \Rightarrow$ segments: $[0, 1)$ and $[1, 4)$. ($T_0 = 1$; $T_1 = 3$). $C = 50$.

Segment $[0, 1)$ ($T = 1$)

- Active event: $[0, 1)$ at village 1 with $p = -2$.
- Village 1: $E = 0$; $L = 2$. $\lfloor \frac{50 \cdot 2}{100} \rfloor = 1$. $R = 1 \cdot (0 + 1) = 1$.
- Other villages: no active events $\Rightarrow$ reward 0.

Segment $[1, 4)$ ($T = 3$)

- At village 2, three events are active simultaneously: $(+5; +4; -6)$. So $E = 5 + 4 = 9$; $L = 6$.
- $\lfloor \frac{50 \cdot 6}{100} \rfloor = 3$. $R = 3 \cdot (9 + 3) = 36$.

Total

- Choose village 1 in $[0, 1)$, then move to village 2 for $[1, 4)$. Movement cost: $\text{dist}(1, 2) = 3$.
- Total reward $= 1 + 36 = 37$, answer $= 37 - 3 = 34$.